Head & Neck Anatomy: Endocrine Glands & Development (70 MCQs)

1. The pituitary gland (hypophysis cerebri) is a vital endocrine organ. Macroscopically, it sits securely inside a depression in the sphenoid bone called the:
a. Sella turcica (hypophyseal fossa)
b. Foramen rotundum
c. Pterygopalatine fossa
d. Lateral pterygoid plate
e. Superior orbital fissure
2. Superiorly, the pituitary gland is covered by a horizontal fold of dura mater that features a central aperture for the infundibulum. This dural fold is the:
a. Falx cerebri
b. Tentorium cerebelli
c. Falx cerebelli
d. Diaphragma sellae
e. Cavum trigeminale
3. The pituitary gland possesses a distinct dual embryological origin. The anterior lobe (adenohypophysis) develops as an upward ectodermal outpocketing from the roof of the primitive oral cavity (stomodeum), known as:
a. Meckel's cartilage
b. Reichert's pouch
c. The thyroglossal tract
d. Rathke's pouch
e. The tuberculum impar
4. The posterior lobe of the pituitary gland (neurohypophysis) shares a completely separate origin. It develops as a downward neuroectodermal growth from the floor of the:
a. Telencephalon
b. Diencephalon (future hypothalamus)
c. Mesencephalon
d. Metencephalon
e. Myelencephalon
5. Histologically, the adenohypophysis is divided into zones. The large, primary functional portion that contains cords of epithelial cells interspersed with fenestrated capillaries is the:
a. Pars intermedia
b. Pars tuberalis
c. Pars nervosa
d. Pars distalis
e. Infundibular stalk
6. Microscopic analysis of the pars distalis reveals cells categorized by their staining properties. The cells that stain intensely with acidic dyes like eosin, responsible for secreting growth hormone (GH) and prolactin (PRL), are the:
a. Chromophobes
b. Basophils
c. Acidophils
d. Melanotrophs

e. Pituicytes

7. Basophils in the anterior pituitary stain blue with basic dyes. Under high magnification, these cells are responsible for synthesizing and releasing:

a. Oxytocin and vasopressin
b. Melatonin purely
c. Thyroid-stimulating hormone (TSH), adrenocorticotropic hormone (ACTH), LH, and FSH
d. Growth hormone only
e. Calcified colloid scales

8. The neurohypophysis (posterior pituitary) does not synthesize hormones. Instead, it histologically consists of unmyelinated nerve fibers, fenestrated capillaries, and modified glial supporting cells called:

a. Astrocytes
b. Pituicytes
c. Oligodendrocytes
d. Microglia
e. Schwann cells

9. The expanded axon terminals inside the posterior pituitary that store oxytocin and antidiuretic hormone (ADH) as dense neurosecretory granules are visible under the microscope as:

a. Nissl bodies
b. Negri bodies
c. Lewy bodies
d. Herring bodies
e. Councilman bodies

10. The vascular connection linking the hypothalamus directly to the anterior pituitary, allowing regulatory releasing factors to reach their target cells quickly, is the:

a. Circle of Willis
b. Pterygoid plexus
c. Cavernous sinus
d. Hypophyseal portal system
e. Internal carotid tract

11. The pineal gland (epiphysis cerebri) is a small, pinecone-shaped structure situated in the midline of the brain. Macroscopically, it projects backward from the posterior wall of the:

a. Lateral ventricle
b. Fourth ventricle
c. Third ventricle
d. Central canal
e. Cisterna magna

12. The pineal gland is an important regulator of circadian rhythms. It contains specialized secretory cells that synthesize the hormone melatonin, known as:

a. Pituicytes
b. Chromaffin cells
c. Podocytes
d. Pinealocytes
e. Purkinje cells

13. A characteristic feature of the pineal gland visible on radiographs of older adults is the presence of calcified mineral deposits, commonly referred to as:
a. Dental pulpal stones
b. Corpora amylacea
c. Brain sand (acervuli/corpora arenacea)
d. Phleboliths
e. Tonsilloliths
14. Developmentally, the pineal gland originates entirely as a midline neuroectodermal evagination from the roof of the caudal portion of the:
a. Diencephalon
b. Telencephalon
c. Myelencephalon
d. Metencephalon
e. Midbrain tectum
15. Let's move to the neck. The thyroid gland is a highly vascular endocrine organ. Macroscopically, it consists of right and left lobes connected across the midline by a horizontal bridge of tissue called the:
a. Piriform recess
b. Isthmus
c. Pyramidal lobe
d. Thyroid prominence
e. Cricothyroid band
16. The horizontal isthmus of the thyroid gland typically crosses the anterior surface of which tracheal structures in a normal subject?
a. The thyroid cartilage plate
b. The cricoid cartilage completely
c. The second, third, and fourth tracheal rings
d. The sixth and seventh tracheal rings
e. The sternal notch margin
17. Frequently, a small, finger-like accessory muscle or fibrous strand extends upward from the thyroid isthmus to attach onto the hyoid bone. This structural variation is known as the:
a. Pyramidal lobe (and levator glandulae thyroideae)
b. Median cervical raphe
c. Thyrohyoid muscle slip
d. Branchial band
e. Stylohyoid ligament remnant
18. The thyroid gland is wrapped in two fascial layers. The true capsule is a dense fibrous layer directly attached to the tissue, while the false capsule is derived from the:
a. Investing layer of deep cervical fascia
b. Pretracheal layer of deep cervical fascia
c. Prevertebral layer
d. Superficial fascial sheet
e. Buccopharyngeal fascia
19. Because the false capsule of the thyroid gland fixes tightly to the cricoid cartilage and thyroid cartilage, the thyroid gland characteristically exhibits which clinical movement?

a. It expands laterally during wide opening of the jaw
b. It moves upward and downward during deglutition (swallowing)
c. It pulsates synchronously with the internal jugular vein
d. It remains completely immobile during all neck movements
e. It retracts backward when the tongue is protruded

20. High-pressure arterial blood is supplied to the thyroid gland via two principal vessels. The superior thyroid artery is a direct branch of the:

a. Common carotid artery
b. Internal carotid artery
c. External carotid artery
d. Thyrocervical trunk
e. Internal thoracic artery

21. The inferior thyroid artery provides extensive blood supply to the lower poles of the gland. This vessel arises as a major branch of the:

a. Thyrocervical trunk (a branch of the subclavian artery)
b. External carotid artery
c. Common carotid artery
d. Internal thoracic artery
e. Vertebral artery

22. In roughly 3% to 10% of individuals, an unmapped accessory artery arises directly from the brachiocephalic trunk or the arch of the aorta to ascend to the thyroid isthmus. This variant vessel is the:

a. Lingual artery
b. Superior laryngeal artery
c. Thyroid ima artery (arteria thyroidea ima)
d. Ascending pharyngeal artery
e. Deep cervical artery

23. Venous drainage of the thyroid gland is mediated by three pairs of veins. The superior and middle thyroid veins empty their contents directly into the:

a. Subclavian vein
b. External jugular vein
c. Internal jugular vein
d. Brachiocephalic vein
e. Anterior jugular venous arch

24. The inferior thyroid veins form an extensive plexus at the lower border of the gland. They drain downwards to empty directly into the:

a. Internal jugular veins
b. Brachiocephalic veins (primarily the left brachiocephalic vein)
c. Subclavian veins
d. External jugular veins
e. Superior vena cava directly

25. During a thyroidectomy, a surgeon must carefully identify a nerve running in close proximity to the superior thyroid artery near the upper pole of the gland. To avoid damaging this nerve, which supplies the cricothyroid muscle, the artery must be ligated close to the gland. This nerve is the:

a. Internal laryngeal nerve
b. Recurrent laryngeal nerve
c. External laryngeal nerve

d. Ansa cervicalis
e. Mylohyoid nerve

26. Near the lower pole of the thyroid gland, the inferior thyroid artery is intimately related to a nerve that ascends in the tracheoesophageal groove to supply the intrinsic muscles of the larynx. This is the:

a. External laryngeal nerve
b. Recurrent laryngeal nerve (branch of CN X)
c. Internal laryngeal nerve
d. Phrenic nerve
e. Cardiac branch of vagus

27. Accidental bilateral injury or transection of the recurrent laryngeal nerves during a thyroidectomy causes complete paralysis of the vocal folds, resulting in a critical emergency characterized by:

a. Loss of general sensation on the tongue
b. Severe hyperacusis
c. Stridor and complete airway obstruction (asphyxiation)
d. Profuse salivation from the parotid duct
e. Sudden drop in systemic blood pressure

28. Histologically, the thyroid gland is composed of thousands of spherical structures called thyroid follicles. These follicles are lined by follicular cells that surround a central lumen filled with a gelatinous material named:

a. Enamel matrix
b. Colloid (containing thyroglobulin)
c. Sebum fluid
d. Perilymph
e. Endolymph matrix

29. When the thyroid gland is highly active, the follicular cells alter their microscopic morphology from a resting flat cuboidal shape into a:

a. Simple squamous profile
b. Stratified squamous sheets
c. Columnar epithelium profile
d. Transitional layer
e. Pseudostratified ciliated sheet

30. Embedded within the follicular epithelium or located in the interfollicular spaces are large, pale-staining parafollicular cells, also known as:

a. Chief cells
b. Oxyphil cells
c. C-cells (which secrete calcitonin)
d. Pituicytes
e. Pinealocytes

31. Calcitonin synthesized by the parafollicular C-cells acts physiologically to lower blood calcium levels. These cells are embryologically derived from the:

a. First pharyngeal pouch
b. Second pharyngeal pouch
c. Ultimobranchial body (derived from the fifth/fourth pharyngeal pouch neural crest)
d. Rathke's pouch ectoderm
e. Stomodeal ceiling

32. The thyroid gland begins its development in the fourth week of gestation as a midline endodermal thickening on the floor of the pharynx, located at a point represented in the adult tongue as the:
a. Sulcus terminalis lateral corner
b. Foramen cecum
c. Lingual frenulum base
d. Sublingual fold
e. Plica fimbriata line
33. As the thyroid primordium descends into the neck anterior to the hyoid bone and laryngeal cartilages, it remains temporarily connected to its site of origin by a narrow epithelial channel called the:
a. Thyroglossal duct
b. Nasolacrimal duct
c. Pterygoid canal
d. Vitelline duct
e. Thymic tract
34. Under normal developmental conditions, the thyroglossal duct undergoes complete degeneration and disappears by the:
a. Sixth week
b. Seventh to eighth week of intrauterine life
c. Time of birth
d. First postnatal year
e. Puberty stage
35. If a segment of the thyroglossal duct fails to obliterate and continues to secrete fluid postnatally, it forms a painless, fluctuating midline swelling in the neck known as a:
a. Branchial cyst
b. Lateral cervical cyst
c. Thyroglossal duct cyst
d. Parotid mucocele
e. Ranula cyst
36. A definitive diagnostic feature of a thyroglossal duct cyst observed during a physical examination is that the mass characteristically moves upward when the patient:
a. Opens their jaw wide or grinds their teeth
b. Swallows or protrudes their tongue
c. Rotates their head to the opposite side
d. Shrugs their shoulders against resistance
e. Takes a deep inspiration
37. If a thyroglossal duct cyst becomes infected and ruptures through the skin of the neck, it forms a permanent pathological tract lined by epithelium, termed a:
a. Branchial fistula
b. Thyroglossal duct fistula (or sinus)
c. Tracheoesophageal tract
d. Oroantral communication
e. Preauricular pit

38. A developmental anomaly occurs when the thyroid gland completely fails to migrate downward from the foramen cecum, leaving a functional mass of thyroid tissue trapped at the base of the tongue. This condition is a:
a. Thyroglossal cyst
b. Lingual thyroid
c. Branchial vestige
d. Persistent Rathke's mass
e. Thymic replacement
39. Small accessory masses of functional thyroid tissue can be left behind anywhere along the normal path of descent. These clusters are formally classified as:
a. Pyramidal adenomas
b. Ectopic thyroid tissue
c. Branchial remnants
d. Thymic cysts
e. Parathyroid rests
40. The parathyroid glands are small, yellowish-brown oval structures. Under normal anatomical conditions, how many total parathyroid glands are present in a human subject?
a. Two glands
b. Three glands
c. Four glands (two superior, two inferior)
d. Six glands
e. Eight glands
41. The superior and inferior parathyroid glands are typically located along which specific anatomical boundary of the thyroid gland?
a. The anterior surface of the isthmus
b. The superior border of the pyramidal lobe
c. The posterior border of the lateral lobes, inside the false capsule
d. Deep inside the tracheal lumen
e. Along the superior nuchal line
42. The superior parathyroid glands are remarkably constant in position. Developmentally, they originate from the endodermal lining of the:
a. First pharyngeal pouch
b. Second pharyngeal pouch
c. Third pharyngeal pouch
d. Fourth pharyngeal pouch
e. Fifth pharyngeal pouch
43. The inferior parathyroid glands exhibit a more variable position. Developmentally, they originate from the endodermal lining of which pharyngeal pouch?
a. First pharyngeal pouch
b. Second pharyngeal pouch
c. Third pharyngeal pouch (alongside the thymus)
d. Fourth pharyngeal pouch exclusively
e. Main stomodeal ectoderm
44. Because the inferior parathyroid glands develop from the same third pharyngeal pouch as the thymus, they migrate downward into the neck with the thymic mass. Consequently, ectopic inferior parathyroid glands can frequently be located within the:

a. Infratemporal fossa roof
b. Superior mediastinum of the thorax (or inside the thymus gland)
c. Submandibular salivary basin
d. Hard palate submucosa
e. Mandibular canal track

45. Histologically, the parathyroid gland parenchyma contains cords of cells supported by a rich capillary network. The most numerous cells, responsible for synthesizing and secreting parathyroid hormone (PTH), are the:

a. Oxyphil cells
b. Parafollicular cells
c. Chief (principal) cells
d. Pituicytes
e. Follicular units

46. Microscopic analysis also reveals a population of larger, highly eosinophilic cells packed with mitochondria that appear in the parathyroid gland around puberty. These cells are the:

a. Chief cells
b. Oxyphil cells
c. C-cells
d. Chromophobes
e. Pinealocytes

47. Parathyroid hormone (PTH) acts to increase blood calcium levels. If all four parathyroid glands are accidentally removed during a radical thyroidectomy, the sudden loss of PTH causes a severe drop in blood calcium, leading to:

a. Congenital goiter
b. Hypocalcemic tetany (characterized by painful muscle spasms and carpopedal spasms)
c. Hypercalcemic crisis
d. Exophthalmos
e. Myxedema coma

48. Two classic clinical signs can elicit latent hypocalcemic tetany. Tapping over the facial nerve in front of the ear triggers a hyper-reflexive twitch of the facial muscles, a phenomenon known as:

a. Trousseau's sign
b. Chvostek's sign
c. Troisier's sign
d. Horner's sign
e. Frey's sign

49. Inflating a blood pressure cuff over the arm blocks arterial flow and triggers a painful carpopedal spasm (flexion of the wrist and thumb with extension of the fingers) within three minutes. This diagnostic indicator is:

a. Chvostek's sign
b. Trousseau's sign
c. Babinski's sign
d. Battle's sign
e. Romberg's sign

50. Pathological hyperfunction of a parathyroid gland (usually due to a benign chief cell adenoma) leads to excessive PTH secretion, causing extensive bone resorption and calcium mobilization. This clinical condition is:

a. Primary hyperparathyroidism
b. Grave's disease
c. Hashimoto's thyroiditis
d. Cretinism
e. Plummer's disease

51. Let's return to the pituitary. The adenohypophysis sends out a thin collar or extension of tissue that wraps around the infundibular stalk like a sleeve. This specific zone is the:

a. Pars distalis
b. Pars intermedia
c. Pars tuberalis
d. Pars nervosa
e. Sella matrix

52. In the fetus, a narrow, cellular zone sits between the pars distalis and the pars nervosa. This zone synthesizes melanocyte-stimulating hormone (MSH) in lower vertebrates and is the:

a. Pars tuberalis
b. Pars intermedia
c. Infundibulum
d. Diaphragma sellae
e. Median eminence

53. Microscopic analysis of the adenohypophysis reveals chromophobes. These cells look pale under the microscope because they contain:

a. Packed zymogen granules
b. Minimal or depleted secretory granules, often representing resting or stem states
c. High concentrations of lipofuscin pigment
d. Calcified sand grains
e. Abundant colloid pools

54. The superior hypophyseal arteries supply the infundibulum and form a primary capillary plexus that collects hypothalamic regulatory factors. These arteries arise directly from the:

a. External carotid artery
b. Maxillary artery
c. Internal carotid artery (cerebral part)
d. Superior thyroid artery
e. Vertebral artery trunk

55. If a developmental remnant of the roof of Rathke's pouch persists inside the pharyngeal wall or sphenoid bone, it can form an accessory mass of anterior pituitary tissue known as a:

a. Pharyngeal hypophysis
b. Lingual pituitary
c. Persistent craniopharyngioma
d. Rathke's cleft cyst
e. Ectopic neurohypophysis

56. A benign, slow-growing tumor that develops from remnants of the epithelial tract of Rathke's pouch along the infundibular stalk, frequently causing visual field deficits due to compression of the optic chiasm, is a:

a. Pituitary adenoma

b. Craniopharyngioma
c. Pinealoblastoma
d. Thyroid Hurthle cell tumor
e. Branchioma

57. Hypersecretion of growth hormone (GH) by an acidophilic adenoma of the anterior pituitary *after* the closure of the epiphyseal plates in adults leads to a classic clinical syndrome characterized by enlargement of the hands, feet, and mandible, known as:
a. Gigantism
b. Acromegaly
c. Cushing's disease
d. Dwarfism
e. Simmonds' disease

58. If the same growth hormone-secreting tumor occurs in a pediatric patient *before* the epiphyseal plates of the long bones fuse, it results in a condition characterized by excessive proportional height, known as:
a. Acromegaly
b. Gigantism
c. Cretinism
d. Marfan syndrome
e. Achondroplasia

59. A patient presents with a pituitary tumor that compresses the central fibers of the optic chiasm situated immediately superior to the diaphragmatic roof. This neural compression produces a classic visual field defect called:
a. Ipsilateral complete blindness
b. Bitemporal hemianopia (loss of both peripheral visual fields)
c. Homonymous hemianopia
d. Central scotoma
e. Quadrantanopia

60. The pineal gland contains supporting interstitial cells interspersed among the pinealocytes. Histologically, these interstitial cells are classified as a type of:
a. Schwann cell
b. Astrocyte (glial cell)
c. Ependymal cell
d. Microglial macro-phage
e. Pericyte unit

61. High-yield vascular patterns of the head specify that the inferior hypophyseal arteries primarily supply which segment of the hypophysis?
a. Pars distalis exclusively
b. Pars tuberalis sleeve
c. Neurohypophysis (pars nervosa)
d. Primary capillary loops of portal bed
e. Optic chiasm border

62. Microscopic evaluation of a thyroglossal duct cyst reveals that its internal wall can be lined by stratified squamous or pseudostratified ciliated columnar epithelium, and its connective tissue capsule frequently contains remnants of:
a. Parathyroid chief cells
b. Functional thyroid follicles or lymphoid aggregates
c. Trapped salivary acini

d. Thymic corpuscles
e. Enamel rods

63. When the thyroid gland undergoes diffuse pathological hyperplasia driven by autoantibodies binding to the TSH receptor, leading to hyperthyroidism and exophthalmos, the condition is diagnosed as:
a. Hashimoto's thyroiditis
b. Grave's disease (exophthalmic goiter)
c. Myxedema
d. Endemic colloid goiter
e. De Quervain's thyroiditis

64. Severe, unmanaged congenital hypothyroidism in infants due to a failure of thyroid gland development or iodine deficiency leads to physical dwarfism and severe mental retardation, a syndrome termed:
a. Myxedema
b. Cretinism
c. Acromegaly
d. Cushing's syndrome
e. DiGeorge syndrome

65. DiGeorge syndrome is a congenital anomaly caused by a deletion on chromosome 22q11.2. It features a complete failure of development of the third and fourth pharyngeal pouches, resulting in a total absence of the:
a. Thyroid gland follicular cells
b. Adenohypophysis and neurohypophysis
c. Parathyroid glands and thymus (causing hypocalcemia and immunodeficiency)
d. Pineal gland and pituitary stalk
e. Submandibular and sublingual salivary glands

66. The superior thyroid vein runs synchronously with the superior thyroid artery. It crosses the carotid sheath to empty its contents directly into the:
a. Subclavian vein
b. External jugular vein
c. Internal jugular vein
d. Left brachiocephalic vein
e. Superior vena cava

67. The middle thyroid vein is short and runs laterally from the lobe of the thyroid gland, crossing the common carotid artery to drain directly into the:
a. External jugular vein
b. Internal jugular vein
c. Subclavian vein
d. Anterior jugular venous arch
e. Innominate trunk

68. The lymphatic vessels from the thyroid isthmus and upper poles track upward to empty primarily into the:
a. Submental lymph nodes
b. Prelaryngeal and upper deep cervical lymph nodes
c. Retropharyngeal lymph nodes
d. Mastoid node groups
e. Superficial cervical chain

69. The lymphatics draining the lower poles of the thyroid gland track downward to target the:

a. Submandibular lymph nodes
b. Pretracheal and paratracheal lymph nodes heading to the deep cervical chain
c. Parotid lymph nodes
d. Retroauricular lymph nodes
e. Submental apex nodes

70. The small lumen or cleft seen histologically within the adenohypophysis, separating the pars distalis from the pars intermedia, represents an embryonic remnant of the original:

a. Infundibular recess cavity
b. Lumen of the third ventricle
c. Lumen of Rathke's pouch (Rathke's cleft)
d. First pharyngeal pouch cleft
e. Thyroglossal tract channel